Equivariant Systems Theory and Observer Design for Autonomous Systems

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Frechet derivative

Any differentiable function $F(E)$,

$$DF(E)[H] = \lim_{t \to 0} \frac{F(E + tH) - F(E)}{t}$$

Examples:
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The derivative of F in the direction of H, is:

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$$\operatorname{grad}(F) = \mathbb{P}_{\mathfrak{so}(3)}(-(I - E)E^\top)E, \quad \text{with} \quad \mathbb{P}_{\mathfrak{so}(3)}(Z) := \frac{1}{2}(Z - Z^\top).$$

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Define

$$F_1(E) = \frac{1}{2}\sum_{i=1}^{n} \|\mathring{y}_i - E\mathring{y}_i\|^2$$

with $\mathring{y}_i \in S^2$, $E \in S0(3)$ such that $\dot{E} = \Delta E$, and $\Delta \in \mathfrak{so}(3)$ $(\Delta^T = -\Delta)$,

Define

$$F_2(H) = \frac{1}{2}\sum_{i=1}^{n} \left\|\mathring{y}_i - \frac{H\mathring{y}_i}{\|H\mathring{y}_i\|}\right\|^2$$

with $\mathring{y}_i \in S^2$, $H \in SL(3)$ such that $\dot{H} = UH$, and $U \in \mathfrak{sl}(3)$ $(\mathrm{tr}(U) = 0)$,

Compute the gradient of F_1 and F_2

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